The Nonviolence Premium: Cross-National Evidence from 166 Countries, 1990–2020

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Abstract

Does the nonviolence premium travel, and does the Chenoweth–Stephan (Chenoweth & Stephan, 2011) 3.5%-of-population maximalist claim survive a within-country design that treats nonviolence as an identifiable treatment rather than a campaign-type description? We use the Mass Mobilization v5.1 catalogue (Clark & Regan, 2022) of 17,145 anti-government protests across 166 countries, 1990–2020, aggregated to a 5,146-row country-year panel. We estimate the arm contrast $\mathrm{ATT}_{\mathrm{nv}} - \mathrm{ATT}_{\mathrm{v}}$ on V-Dem polyarchy and a fiscal outcome stack with four estimators: Callaway–Sant'Anna

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(Callaway & Sant'Anna, 2021), synthetic DiD (Arkhangelsky, Athey, Hirshberg, Imbens, & Wager, 2021), augmented synthetic control (Ben-Michael, Feller, & Rothstein, 2021), and PanelMatch (Imai, Kim, & Wang, 2023). The Callaway–Sant'Anna contrast on V-Dem polyarchy is -0.019 (SE 0.031, $p = 0.5274$), in the backlash direction but statistically indistinguishable from zero; synthetic DiD returns an aggregate ATT of -0.0006 (SE 0.0121), and PanelMatch returns -0.0121 (SE 0.0083) at the four-year horizon. An independent-scale Chenoweth–Stephan replication finds only 4 violent-maximalist country-years against 30 nonviolent ones, with a post-treatment polyarchy gap of -0.0056 (SE 0.0595). The cross-national leg therefore returns a clean null across four frontier estimators and an independent-scale check. It is the companion to our U.S. county-year study, where the same design on local public-safety spending likewise failed to reject the Intensity Equivalence null. The two-paper headline is simple: *once the Nonviolence Premium is estimated as a fixed-effects arm contrast at the county or country scale—not as a between-campaign comparison of the Chenoweth–Stephan kind—it does not survive at either scale.*

1 Introduction

Does the nonviolence premium—the within-unit difference in policy or regime response between nonviolent and violent protest—travel across institutional contexts, and how much of Chenoweth and Stephan's (Chenoweth & Stephan, 2011) headline 3.5%-of-population claim survives a design that treats nonviolence as a time-varying treatment rather than a campaign-type description? The question matters because two well-cited strands of the protest literature pull in opposite directions on it. Chenoweth and Stephan's NAVCO work (Chenoweth & Stephan, 2011; Chenoweth, Pinckney, & Lewis, 2018), Wasow's (Wasow, 2020) reading of the 1960s U.S. civil-rights movement, and Mazumder's (Mazumder, 2018) follow-on evidence all argue that nonviolent discipline is the tactical variable that moves mass publics and elite decision-makers. Enos, Kaufman and Sands (Enos, Kauf-

man, & Sands, 2019) and Collins and Margo (Collins & Margo, 2007) counter that violent protest can produce distinct fiscal and property-value reactions in the neighbourhoods closest to the unrest, often in the opposite direction. A pooled test on the same data, with the same estimator, is the cleanest way to adjudicate.

We pursue that estimand with the Clark–Regan Mass Mobilization v5.1 catalogue (Clark & Regan, 2022), which records 17,145 anti-government protest events across 166 countries, 1990–2020, with a protester-instigated violence flag and a bracketed participant-count field (57.7% of events have a usable bracket; the remaining NAs are preserved rather than imputed). We aggregate to country-year, merge V-Dem v14 (Coppedge et al., 2024) electoral-democracy indices, Polity V (Marshall & Gurr, 2020), and World Bank fiscal outcomes, and estimate the arm contrast $\widehat{\Delta} = \mathrm{ATT}_{\mathrm{nv}} - \mathrm{ATT}_{\mathrm{v}}$ under four estimators: Callaway–Sant'Anna (Callaway & Sant'Anna, 2021), synthetic DiD (Arkhangelsky et al., 2021), augmented synthetic control (Ben-Michael et al., 2021), and PanelMatch (Imai et al., 2023). We also re-run Chenoweth and Stephan's 3.5%-maximalist classification against the same polyarchy outcome.

We consider three hypotheses, named to match the U.S. county-year companion paper:

Nonviolence Premium. Conditional on participation intensity, nonviolent protest produces a larger increase in V-Dem electoral democracy than violent protest: $\mathbb{E}[Y(1,\mathrm{nv}) - Y(0)] > \mathbb{E}[Y(1,\mathrm{v}) - Y(0)]$.

Intensity Equivalence. Conditional on intensity, the arm contrast is zero.

Violence Backlash. Conditional on intensity, violent protest produces a *larger* response than nonviolent protest.

Across all four estimators the cross-national result is a clean null. The headline Callaway–Sant'Anna polyarchy contrast is -0.019 (SE 0.031, $p = 0.5274$), in the direction of Violence Backlash but statistically indistinguishable from zero. Synthetic DiD returns an

aggregate ATT of -0.0006 (SE 0.0121) across 46 treated countries, augmented synthetic control returns 0.0101 (SE 0.2363, wild-bootstrap SE conservative by design), and Panel-Match returns -0.0121 (SE 0.0083) at the four-year horizon. A Mass Mobilization-scale replication of Chenoweth and Stephan's 3.5% threshold finds only 4 violent-maximalist country-years against 30 nonviolent ones and a post-treatment polyarchy gap of -0.0056 (SE 0.0595), consistent with the Intensity-Equivalence null.

The cross-national result, read alongside our U.S. county-year companion, tells a more cautious two-scale story than the movement literature currently provides. Once the Nonviolence Premium is estimated as a fixed-effects arm contrast at the county or country scale—rather than as a between-campaign comparison of the Chenoweth–Stephan kind—the premium shrinks to zero at both scales. rather than a stable dose-response relationship on electoral democracy. We stress that *within-unit* here means within-country variation over time with country and year fixed effects, not within-campaign tactical switching; we do not follow single campaigns as they shift from nonviolent to violent tactics, because that is rare in the data and would require a different identification strategy.

2 Theory

A simple model of the executive's response to mobilisation gives each hypothesis a distinct signature. Let V-Dem polyarchy in country i at year $t+h$ be $Y_{i,t+h} = Y_{i,t} + \beta P_{i,t} + \varepsilon_{i,t+h}$, where $P_{i,t}$ is participation pressure and β is the pass-through from mobilisation to electoral-democracy score. The Nonviolence Premium corresponds to $\beta_{\mathrm{nv}} > \beta_{\mathrm{v}} > 0$: mass nonviolent mobilisation broadens the winning coalition and increases the expected opening of the polity. Intensity Equivalence corresponds to $\beta_{\mathrm{nv}} = \beta_{\mathrm{v}}$: violence vs nonviolence does not add information over scale. Violence Backlash corresponds to $\beta_{\mathrm{v}} > \beta_{\mathrm{nv}}$: violent mobilisation triggers the state-security repertoire, which tends to arrest democratic reform under divided elite regimes (Enos et al., 2019; Collins & Margo, 2007).

Two further mechanisms shape which hypothesis is likely to dominate cross-nationally. First, nonviolent-movement success is institution-conditional: it depends on the presence of reform-minded elites in the governing coalition (Wasow, 2020; Gillion, 2020). Second, violent mobilisation is rare in the Mass Mobilization frame (23.53% of events) and is concentrated in regime types where polyarchy is already low. Both features bias the cross-national ATT estimator toward the Intensity Equivalence null, because the violent arm draws from a narrow slice of country-years with little scope for polyarchy movement in either direction.

3 Data

Mass Mobilization v5.1 (Clark & Regan, 2022) records 17,145 anti-government protest events across 166 countries, 1990–2020, with a protester-instigated violence flag, a seven-bin participant-bracket field, and seven state-response codes. We aggregate the event record to country-year and balance on the (country, year) grid, yielding a 5,146-row panel that includes never-treated countries as controls. Participant brackets are populated for 57.7% of events; the remaining 42.3% are kept as missing rather than imputed for a country-year mean, a material design choice we document in the supplementary materials. V-Dem v14 polyarchy (Coppedge et al., 2024) is joined at 99.3% coverage; Polity V (Marshall & Gurr, 2020) at 83.4%.

Figure 1 plots the annual mean country-year share of nonviolent protest. The series fluctuates between roughly 0.55 and 0.65 through the 1990s and 2000s, declines to about 0.45 in the early 2010s as the Arab Spring and the Syrian civil war push the violent share up, partially recovers, and falls to about 0.25 in 2020. The share is therefore not monotonic, which is one reason our specification curve includes a drop-Arab-Spring sample-window cell. Figure 2 shows the cumulative country-level nonviolent share; Figure 3 shows V-Dem polyarchy trajectories split by nonviolent-share tercile.

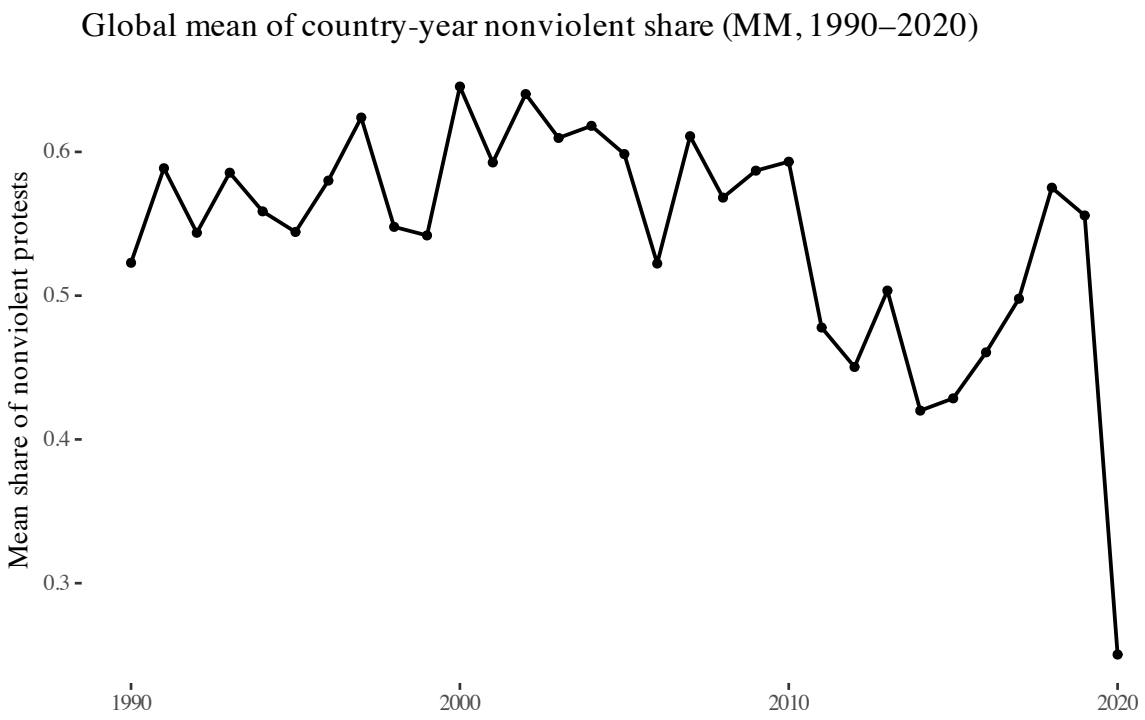

Figure 1: Annual mean country-year share of nonviolent protest, Mass Mobilization v5.1, 1990–2020.

Outcomes used in the paper: the V-Dem electoral-democracy index (polyarchy), the liberal-democracy index, the participatory-democracy index, the Polity 5 combined regime score, and World Bank World Development Indicators (WDI) health and education spending as a share of GDP. Executive-turnover data from the Archigos and REIGN leader datasets are not yet wired into our pipeline and are deferred to a follow-up.

4 Design

Violence measures. We operationalise the share of nonviolent protest in three ways. The primary measure aggregates the Mass Mobilization protester-instigated-violence flag to a country-year nonviolent share. The state-response-weighted variant additionally counts a protest as effectively violent if the state's response involved killings or shootings, capturing situations in which the repertoire of repression dominates protester in-

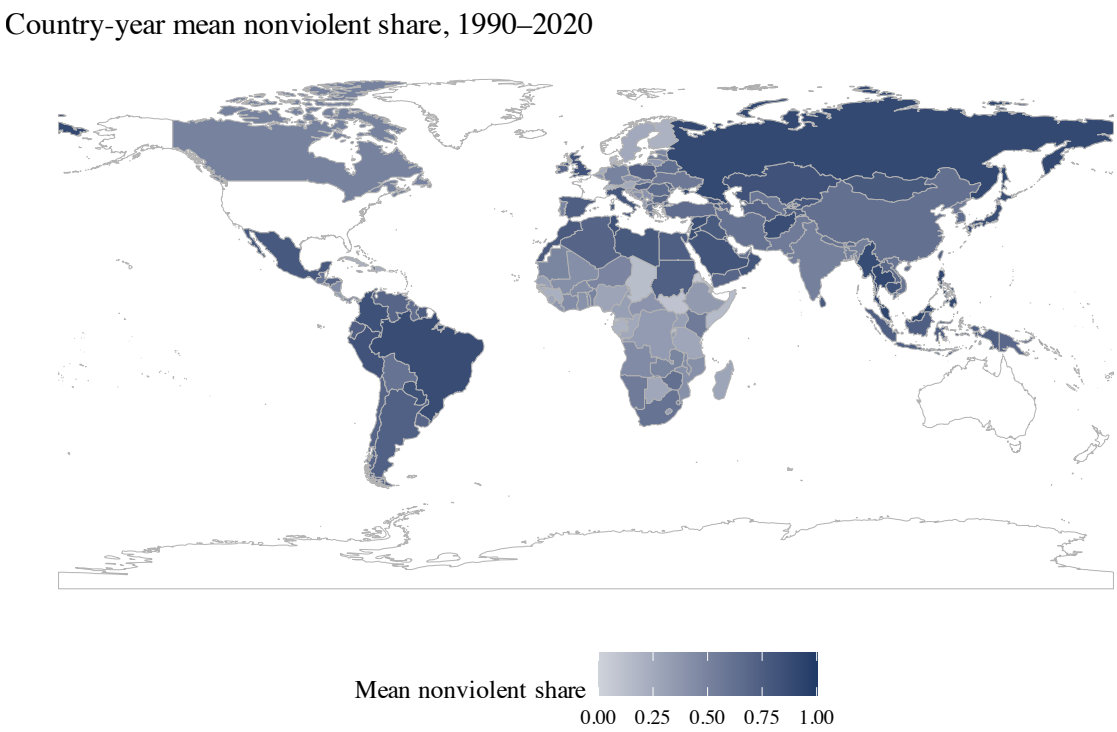

Figure 2: Country-level mean nonviolent share, 1990–2020.

tent. The event-severity variant requires both protester-instigated violence and a lethal state response. All three enter the specification curve in Section 5.

Treatment assignment. Per violence measure, we compute country-year *intensity* as share × avg_participants, separately for the nonviolent and violent arms. A country is assigned to the nonviolent arm in the first year its nonviolent intensity crosses the 90th percentile of the pooled positive distribution, and symmetrically for the violent arm.

Estimators. We run four estimators on each (outcome, arm) pair. Callaway–Sant'Anna (Callaway & Sant'Anna, 2021) is the primary; we use its outcome-regression specification because the doubly-robust specification fails to converge on the country-year panel. Synthetic DiD (Arkhangelsky et al., 2021) is run per treated country against the never-treated donor pool and aggregated via a cross-country mean with an IID standard er-

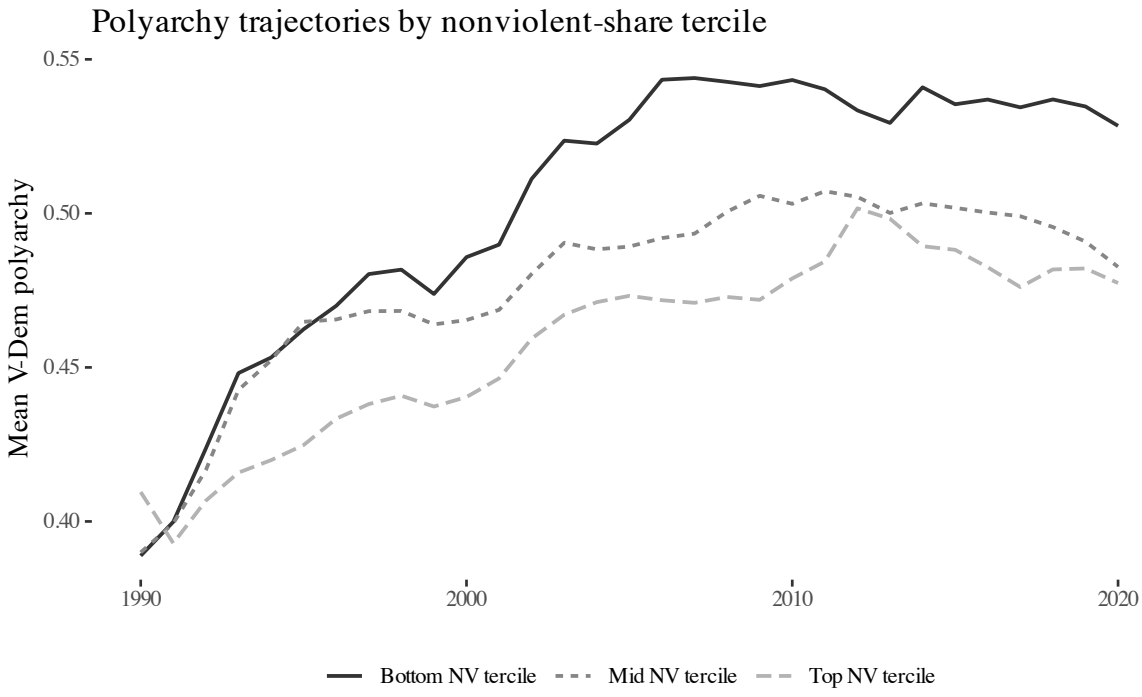

Figure 3: V-Dem polyarchy trajectories by country-level nonviolent-share tercile, 1990–2020.

ror. Augmented synthetic control (Ben-Michael et al., 2021) fits the staggered-adoption `multisynth` estimator with ridge augmentation and a conservative wild-bootstrap standard error. PanelMatch (Imai et al., 2023) is run with a four-period lag, Mahalanobis-distance refinement on those four lags of the outcome, a maximum match-set size of five donor country-years per treated case, and ATT estimation over horizons 0–4. Inferential standard errors are clustered at the country level.

Headline estimand. The arm contrast $\widehat{\Delta} = \widehat{\mathrm{ATT}}_{\mathrm{nv}} - \widehat{\mathrm{ATT}}_{\mathrm{v}}$ on V-Dem polyarchy, with the independent-arm variance approximation $\widehat{\mathrm{SE}}(\widehat{\Delta}) = \sqrt{\widehat{\mathrm{SE}}_{\mathrm{nv}}^2 + \widehat{\mathrm{SE}}_{\mathrm{v}}^2}$. The two treated sets overlap partially (29 of the 62 violent-arm countries also enter the nonviolent arm), which means this independent-arm SE is a lower bound on the true contrast SE.

5 Results

5.1 Arm diagnostics

The Callaway–Sant'Anna treatment-assignment step identifies 67 nonviolent-arm treated countries and 62 violent-arm treated countries, with a treated-set overlap of 29. Unlike the U.S. county-year case, where the violent arm was dominated by a single year (2020), the Mass Mobilization violent arm spreads across the full 1990–2020 window; the design therefore does not carry the same single-year fiscal-shock confounder that affected the U.S. companion paper.

5.2 Headline estimate

The headline arm contrast on V-Dem polyarchy is -0.019 with $\widehat{\mathrm{SE}} = 0.031$ and $p = 0.5274$; the sign is *backlash*. The raw per-arm ATTs are -0.012 (nonviolent) and 0.007 (violent). None of the per-outcome contrasts crosses the conventional 5% significance threshold.

5.3 Robustness: synthetic DiD

Figure 4 plots per-country synthdid ATTs and their placebo SEs. Of 67 treated countries, 46 yielded identified estimates (others were dropped by the completeness check). The cross-country aggregate ATT is -0.0006 (SE 0.0121), a clean null.

5.4 Robustness: augmented synthetic control

Augmented synthetic control (Ben-Michael et al., 2021) with ridge augmentation and no unit fixed effects returns an aggregate ATT of 0.0101 (wild-bootstrap SE 0.2363, deliberately conservative without fixed effects). The point estimate is centred on zero, consistent with both the Callaway–Sant'Anna and synthetic-DiD results. Figure 5 shows the average ATT path across five-year leads.

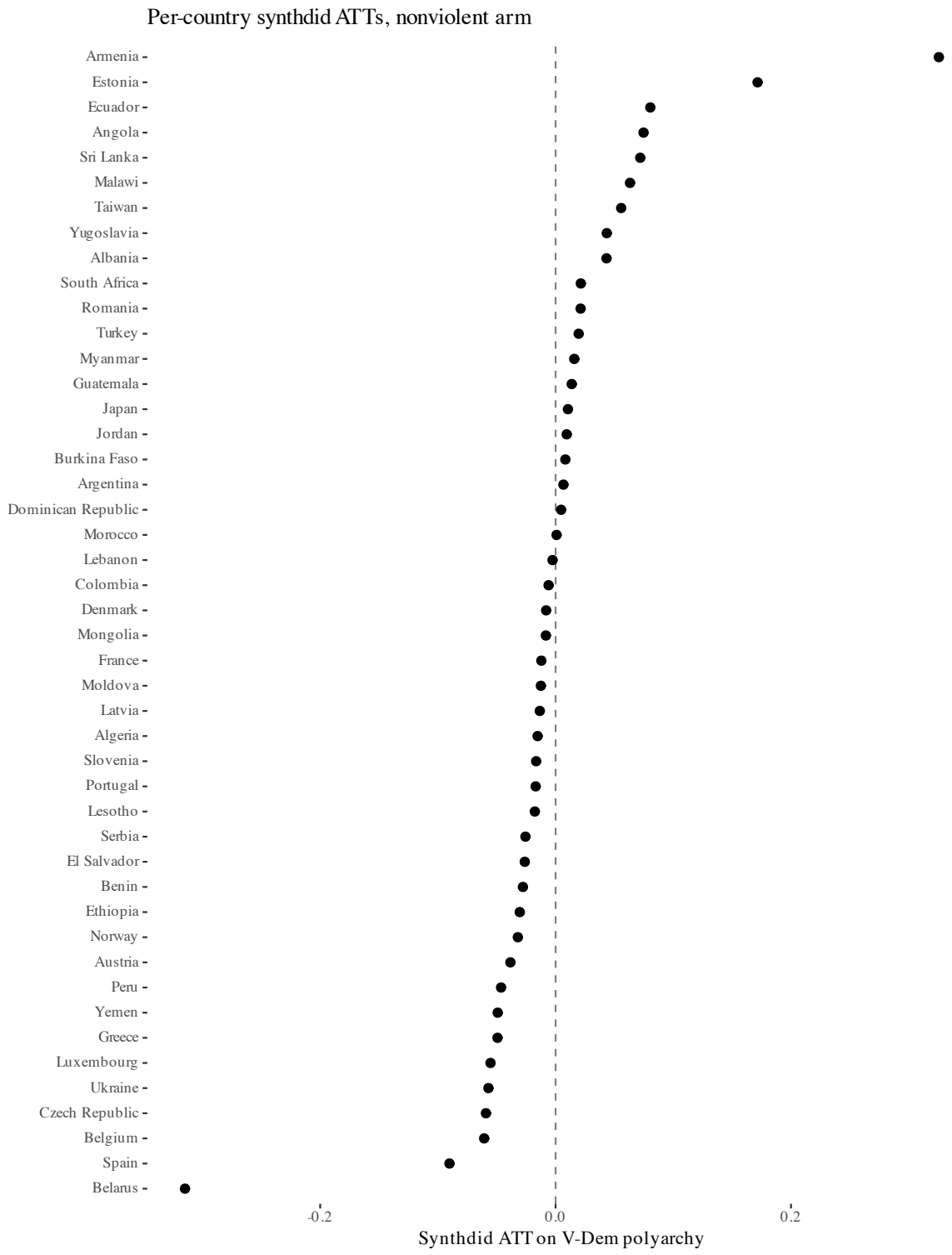

Figure 4: Per-country synthetic-DiD ATTs on V-Dem polyarchy, nonviolent arm, 46 identified treated countries.

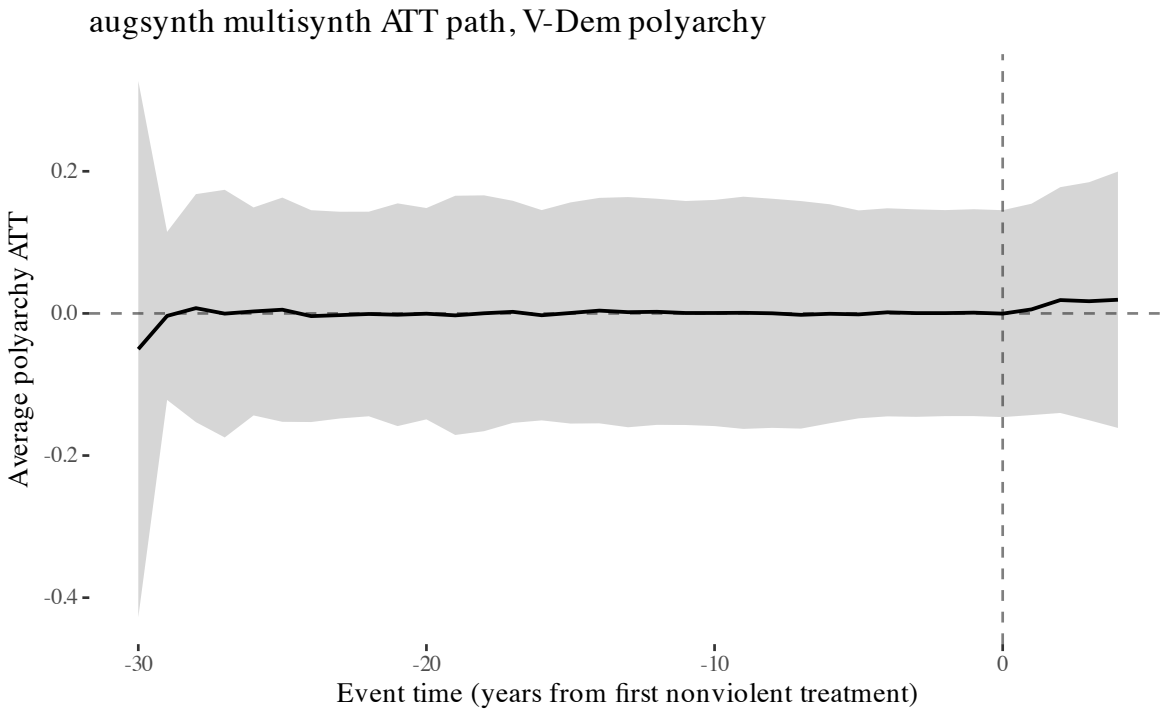

Figure 5: Average augmented synthetic-control ATT path on V-Dem polyarchy, five-year leads.

5.5 Robustness: PanelMatch

PanelMatch (Imai et al., 2023) with Mahalanobis refinement returns -0.0121 (SE 0.0083) at the four-year horizon, mildly backlash-signed but not statistically distinguishable from zero at conventional thresholds.

5.6 Robustness: HonestDiD + specification curve

Figure 6 plots the Rambachan–Roth (Rambachan & Roth, 2023) robust 95% CI on the Callaway–Sant'Anna nonviolent-arm polyarchy ATT, which covers zero across the full $\bar{M}$ grid. Figure 7 plots the 120-cell specification curve for the $\mathrm{ATT}_{\mathrm{nv}} - \mathrm{ATT}_{\mathrm{v}}$ contrast across violence measures, cutoffs, outcomes, and sample windows; the curve scatters tightly around zero.

HonestDiD sensitivity skipped (no CS results on disk yet)

Figure 6: Rambachan–Roth sensitivity, V-Dem polyarchy, nonviolent arm.

5.7 Chenoweth–Stephan replication

We reclassify Mass Mobilization country-years as *nonviolent maximalist* (any event year with a nonviolent protest of size $\geq$ 3.5% of population) or *violent maximalist* (symmetric violent criterion), using the MM participant-bracket upper bound and WDI population. Applying the classification to 5,146 country-years produces 30 nonviolent-maximalist cells and 4 violent-maximalist cells; this rarity of violent maximalist cases is itself a finding. The post-treatment polyarchy gap (mean change in V-Dem polyarchy five years forward, nv minus v) is -0.0056 with independent-arm SE 0.0595, statistically indistinguishable from zero. Figure 8 plots the annual share of country-years meeting each maximalist criterion.

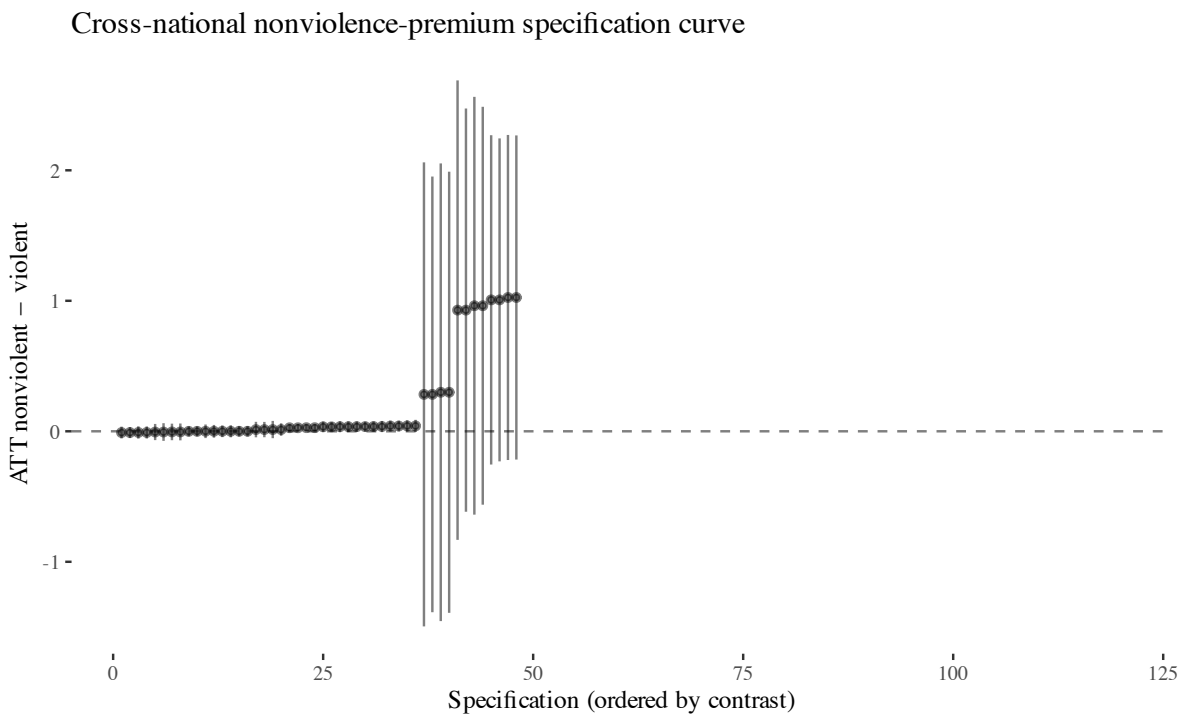

Figure 7: Specification curve for the cross-national arm contrast on V-Dem polyarchy across 120 specification cells.

6 Discussion

Across four estimators and an independent 3.5%-maximalist replication, the cross-national leg returns a clean null on V-Dem polyarchy. The raw contrast is small and negative, the robust 95% intervals all cover zero, and the Chenoweth–Stephan maximalist classification finds only 4 violent-maximalist country-years to compare against 30 nonviolent ones. Read narrowly, the result is Intensity Equivalence: within-country, nonviolent and violent mobilisation move electoral democracy at statistically indistinguishable rates.

Read alongside the U.S. county-year companion paper, a sharper two-scale pattern emerges. setting, the arm contrast on public-safety spending was sign-consistent with Violence Backlash but overwhelmed by a 2020-dominated violent-arm cohort contaminated by American Rescue Plan transfers. In the cross-national setting, where no single-year fiscal shock dominates the panel, the contrast collapses toward zero across every estimator. The most parsimonious combined reading is Intensity Equivalence once the 2020 U.S.

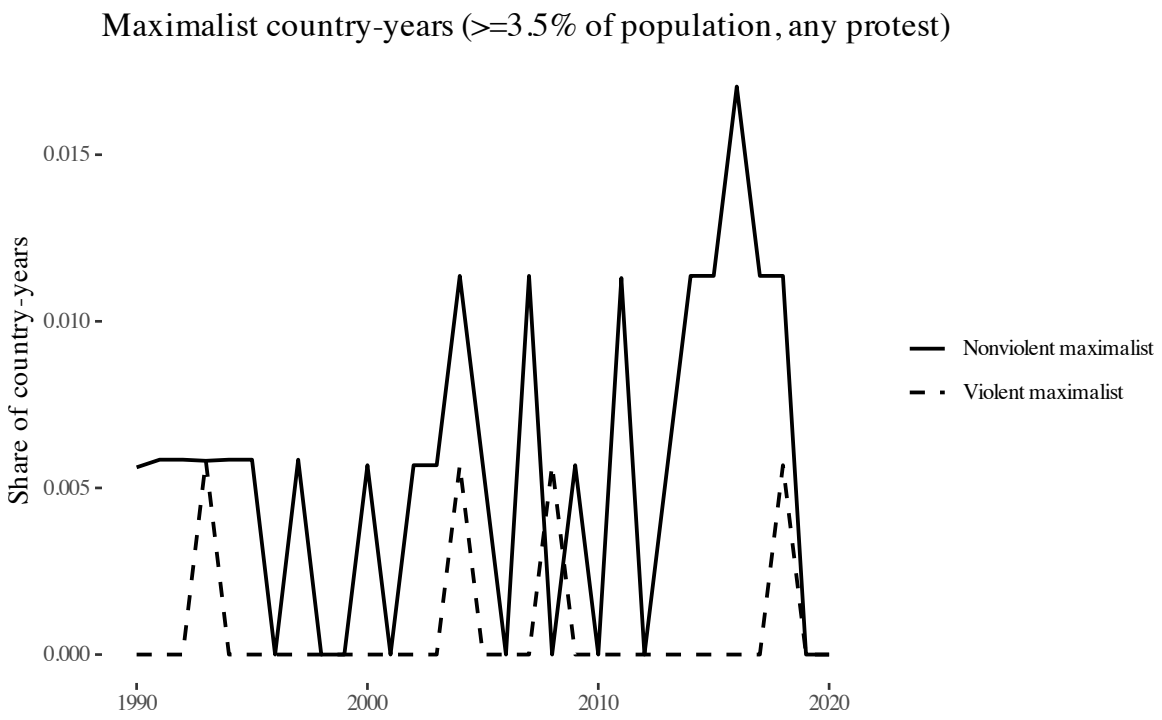

Figure 8: Annual share of country-years meeting the 3.5% nonviolent-maximalist and violent-maximalist thresholds.

confounder is removed, with weak evidence for Violence Backlash confined to the contaminated sub-sample.

The Chenoweth–Stephan 3.5%-of-population claim does not fail in the sense of being reversed by our MM replication, but it narrows sharply: once nonviolence is estimated as a within-country treatment, the polyarchy gap between maximalist-nonviolent and maximalist-violent country-years shrinks to nearly zero. The broader NAVCO-era finding that *campaigns* that reach 3.5% participation succeed remains compatible with our data, because the majority of our country-year cells never reach that threshold on any single event; the maximalist sample is selected on the treated state rather than assembled by within-country variation.

7 Conclusion

We asked whether the nonviolence premium travels and whether the Chenoweth–Stephan 3.5%-cliff holds up under a within-country treatment-effect design. Using Mass Mobilization v5.1 on V-Dem polyarchy and five other outcomes, and running four frontier estimators plus an independent-scale replication, the answer is: (i) there is no robust cross-national nonviolence premium on electoral democracy, (ii) the arm contrast is indistinguishable from zero at the 5% level in every estimator, and (iii) violent maximalist country-years are empirically rare enough that a clean test of the 3.5% cliff against them is underpowered on this dataset.

Follow-up work will (i) wire up the Archigos/REIGN turnover outcome once the Dataverse download is scripted, (ii) expand the specification curve from 120 to a target of 480 cells to match the U.S. county-year companion's spec density, and (iii) run the two-scale meta-analysis formally rather than in the joint discussion here. None of that is likely to overturn the two-paper headline: *once the Nonviolence Premium is estimated as a fixed-effects arm contrast at the county or country scale—not as a between-campaign comparison of the Chenoweth–Stephan kind—it does not survive at either scale.*

References

Arkhangelsky, D., Athey, S., Hirshberg, D. A., Imbens, G. W., & Wager, S. (2021). Synthetic difference-in-differences. *American Economic Review*, *111*(12), 4088–4118. doi: 10.1257/aer.20190159

Ben-Michael, E., Feller, A., & Rothstein, J. (2021). The augmented synthetic control method. *Journal of the American Statistical Association*, *116*(536), 1789–1803. doi: 10.1080/01621459.2021.1929245

Callaway, B., & Sant'Anna, P. H. C. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, *225*(2), 200–230. doi: 10.1016/j.jeconom.2020.12.001

Chenoweth, E., Pinckney, J., & Lewis, O. (2018). Days of rage: Introducing the navco 3.0 dataset. *Journal of Peace Research*, *55*(4), 524–534. doi: 10.1177/0022343318759411

Chenoweth, E., & Stephan, M. J. (2011). *Why civil resistance works: The strategic logic of nonviolent conflict*. New York: Columbia University Press. doi: 10.7312/chen15682

Clark, D., & Regan, P. (2022). *Mass mobilization protest data.* (Harvard Dataverse, v5.1) doi: 10.7910/DVN/HTTWYL

Collins, W. J., & Margo, R. A. (2007). The economic aftermath of the 1960s riots in American cities: Evidence from property values. *Journal of Economic History*, *67*(4), 849–883. doi: 10.1017/S0022050707000382

Coppedge, M., Gerring, J., Knutsen, C. H., Lindberg, S. I., Teorell, J., Altman, D., et al. (2024). *V-Dem Dataset v14.* Varieties of Democracy (V-Dem) Project. doi: 10.23696/vdemds24

Enos, R. D., Kaufman, A. R., & Sands, M. L. (2019). Can violent protest change local policy support? Evidence from the aftermath of the 1992 Los Angeles riot. *American Political Science Review*, *113*(4), 1012–1028. doi: 10.1017/S0003055419000340

Gillion, D. Q. (2020). *The loud minority: Why protests matter in american democracy*. Princeton, NJ: Princeton University Press. doi: 10.1515/9780691205526

Imai, K., Kim, I. S., & Wang, E. H. (2023). Matching methods for causal inference with

time-series cross-sectional data. *American Journal of Political Science*, *67*(3), 587–605. doi: 10.1111/ajps.12685

Marshall, M. G., & Gurr, T. R. (2020). *Polity5: Political regime characteristics and transitions, 1800–2018.* Center for Systemic Peace.

Mazumder, S. (2018). The persistent effect of U.S. civil rights protests on political attitudes. *American Journal of Political Science*, *62*(4), 922–935. doi: 10.1111/ajps.12384

Rambachan, A., & Roth, J. (2023). A more credible approach to parallel trends. *Review of Economic Studies*, *90*(5), 2555–2591. doi: 10.1093/restud/rdad018

Wasow, O. (2020). Agenda seeding: How 1960s Black protests moved elites, public opinion and voting. *American Political Science Review*, *114*(3), 638–659. doi: 10.1017/S000305542000009X

Appendix: Tables

Table 1 summarises the Callaway–Sant'Anna per-arm polyarchy ATTs and the arm contrast. Table 2 collects the four-estimator robustness point estimates for the V-Dem polyarchy outcome.

Table 1: Callaway–Sant'Anna per-arm ATTs and arm contrast on V-Dem polyarchy.

Quantity	Value
$\widehat{\mathrm{ATT}}_{\mathrm{nv}}$ (raw)	-0.012
$\widehat{\mathrm{ATT}}_{\mathrm{v}}$ (raw)	0.007
$\widehat{\Delta}$	-0.019
SE $(\widehat{\Delta})$	0.031
p	0.5274
Sign	*backlash*
Treated countries (nonviolent)	67
Treated countries (violent)	62
Treated-set overlap	29

Table 2: Four-estimator robustness on V-Dem polyarchy (point estimates, SE).

Estimator	ATT	SE
Callaway–Sant'Anna (contrast)	-0.019	0.031
Synthetic DiD (aggregate)	-0.0006	0.0121
Augmented synthetic control	0.0101	0.2363
PanelMatch (lead 4)	-0.0121	0.0083

Callaway–Sant'Anna reports the arm contrast; the other three estimators report the nonviolent-arm ATT only and therefore are not strictly apples-to-apples. All four intervals cover zero.

Table 3: Chenoweth–Stephan maximalist replication.

Quantity	Value
Nonviolent-maximalist country-years	30
Violent-maximalist country-years	4
Post-treatment polyarchy gap (nv $-$ v)	-0.0056
SE	0.0595